PRACTICE 1 - EXAMPLES OF A QUANTITATIVE RESEARCH

Analysing the Changes in the Media and Communication Market by Quantitative Methods

Global and Vietnam Evidence, 2021-2025

Course: _____

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Quantitative market report | APA 7th edition referencing

Executive Summary

This report uses quantitative methods to examine the rapid restructuring of media and communication markets globally and in Vietnam during 2021-2025. The analysis combines official connectivity statistics, platform financial reports, audience-reach data, Vietnamese media-sector statistics, and commercial advertising-market estimates. The main techniques are trend analysis, year-on-year change, compound annual growth rate (CAGR), revenue indexing, market-share evidence, and descriptive distribution analysis.

Indicator	Start	Latest	Quantitative finding
Vietnam internet users	68.72m (2021)	79.80m (2025)	+16.1%; 3.8% CAGR
World internet users	4.9bn (2021)	6.0bn (2025)	+22.4%; 5.2% CAGR
Vietnam digital ad market	US$903m (2021)	US$1.47bn (2025 est.)	+62.8%; 13.0% CAGR
Press total revenue	VND9,500bn (2022)	VND8,080bn (2024)	-14.9%
Radio/TV total revenue	VND15,158bn (2022)	VND9,140bn (2024)	-39.7%

Main interpretation

The numbers point to a clear reallocation of attention and advertising value from traditional media toward digital platforms. Vietnam added more than 11 million internet users between 2021 and 2025, while the estimated digital advertising market expanded by almost two-thirds. At the same time, press and broadcasting revenues contracted sharply in 2022-2024. Platform economics reinforce this shift: Google, Meta and Amazon all expanded advertising revenues over 2021-2025, while Apple enlarged a high-value services ecosystem that includes advertising, digital content, cloud, payments and app distribution.

The transition is not uniform. In the 2024 digital-transformation maturity assessment, only 9.9% of 282 Vietnamese press agencies were rated “excellent,” while 38.7% were rated “poor.” Thus, the central finding is not that traditional media is disappearing, but that its economic model is being rebuilt around digital distribution, data, multimedia formats, platform dependence and diversified revenue.

Research value for the final assessment

For a graded research paper, the strongest defensible argument is that Vietnam’s media market has changed structurally rather than cyclically. The evidence shows simultaneous audience digitalization, rapid digital-ad growth, concentration of platform power, revenue pressure on legacy media, and uneven digital capability across domestic news organizations. These indicators can be used as the empirical background for a final assessment on media economics, platformization, digital transformation or advertising-market change.

1. Research Background, Questions and Method

1.1 Background

Media markets have historically linked content producers, audiences and advertisers. Digital platforms alter this relationship because they combine distribution, identity, data, recommendation systems, payment infrastructure and advertising exchanges in a single ecosystem. This makes audience size alone insufficient for explaining market power: control over discovery, targeting, measurement, app distribution and commerce data becomes equally important.

Vietnam is a particularly useful case because connectivity is already high, social media reaches most internet users, e-commerce is expanding, and the state has explicitly promoted digital transformation of press agencies. The period 2021-2025 captures the post-pandemic acceleration of online consumption and the subsequent adjustment of traditional media revenues.

1.2 Research questions

- RQ1: How did the number and penetration of internet users change globally and in Vietnam from 2021 to 2025?
- RQ2: What do revenue and audience indicators show about the role of Google, Amazon, Facebook/Meta and Apple (GAFA) in media and communication markets?
- RQ3: How has the structure and digital maturity of Vietnamese press and broadcasting changed?
- RQ4: How rapidly has digital advertising expanded in Vietnam, and how does this compare with traditional media revenue trends?
- RQ5: Do the combined indicators support the proposition that Vietnam’s media and communication market changed rapidly during the last five years?

1.3 Quantitative methods

Method	Formula / operationalization	Use in this report
Absolute change	Latest value - initial value	Internet users; revenue
Percentage change	(Latest / initial - 1) x 100	Market expansion/contraction
CAGR	(Latest / initial)^(1/n) - 1	Multi-year growth speed
Index number	Year value / base value x 100	Compare differently sized firms/markets
Distribution share	Category count / total x 100	Digital maturity ratings
Reach ratio	Platform ad reach / internet users	Platform penetration in Vietnam

1.4 Data quality and limitations

The report distinguishes official statistics from modeled market estimates. ITU and Vietnamese government figures are treated as official or administrative sources. DataReportal figures are compiled from platform advertising tools and other sources and are best interpreted as audience-reach estimates rather than unique active users. The Vietnam digital-advertising series is a commercial market estimate, not an official national-account measure. GAFA figures use company-reported revenues; Apple Services is broader than advertising and is therefore analyzed as a platform-monetization indicator rather than directly added to advertising totals.

2. Quantitative Analysis of Internet Users: World and Vietnam

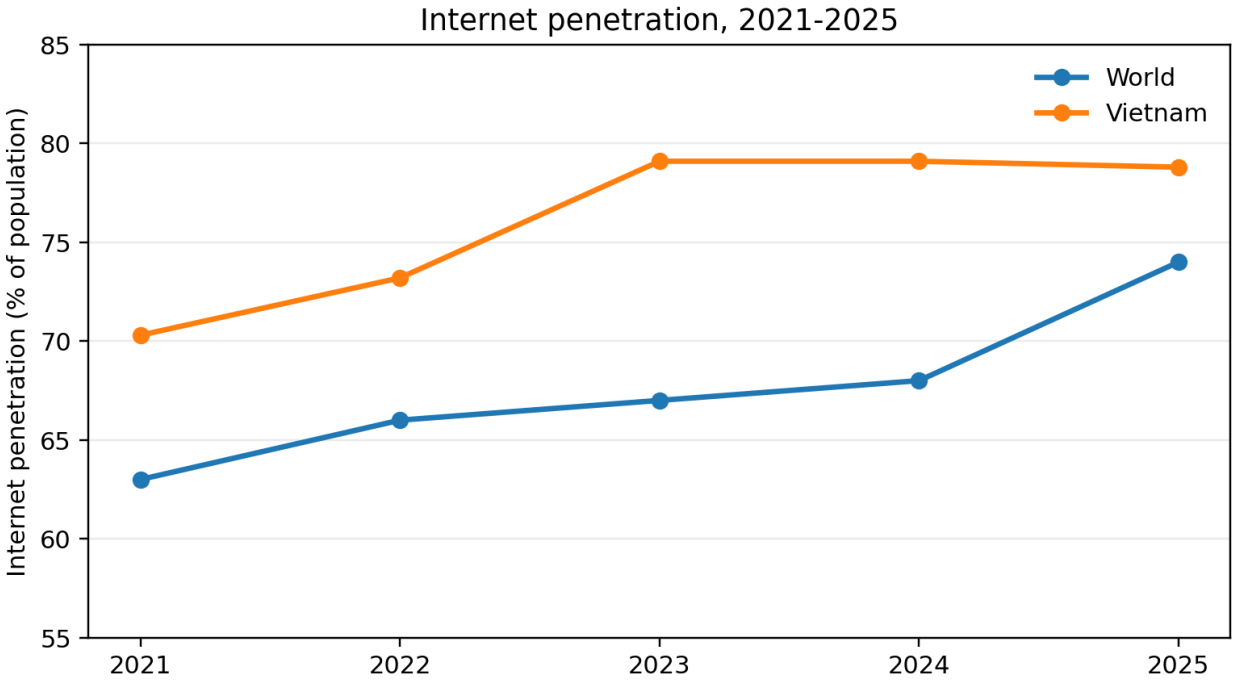


Figure 1. Internet penetration in the world and Vietnam, 2021-2025.

Source: ITU (2021-2025) and DataReportal (2021-2025).

Vietnam remained above the global internet penetration rate throughout the period. Its reported penetration increased from 70.3% in 2021 to 78.8% in 2025, an 8.5 percentage-point increase. The global published estimate rose from 63% to 74%, an 11-point increase. The small decline in Vietnam's penetration from 79.1% in 2024 to 78.8% in 2025 should not be interpreted as a collapse in use: the absolute user count still increased, while population estimates and data revisions affect penetration ratios (Kemp, 2024, 2025).

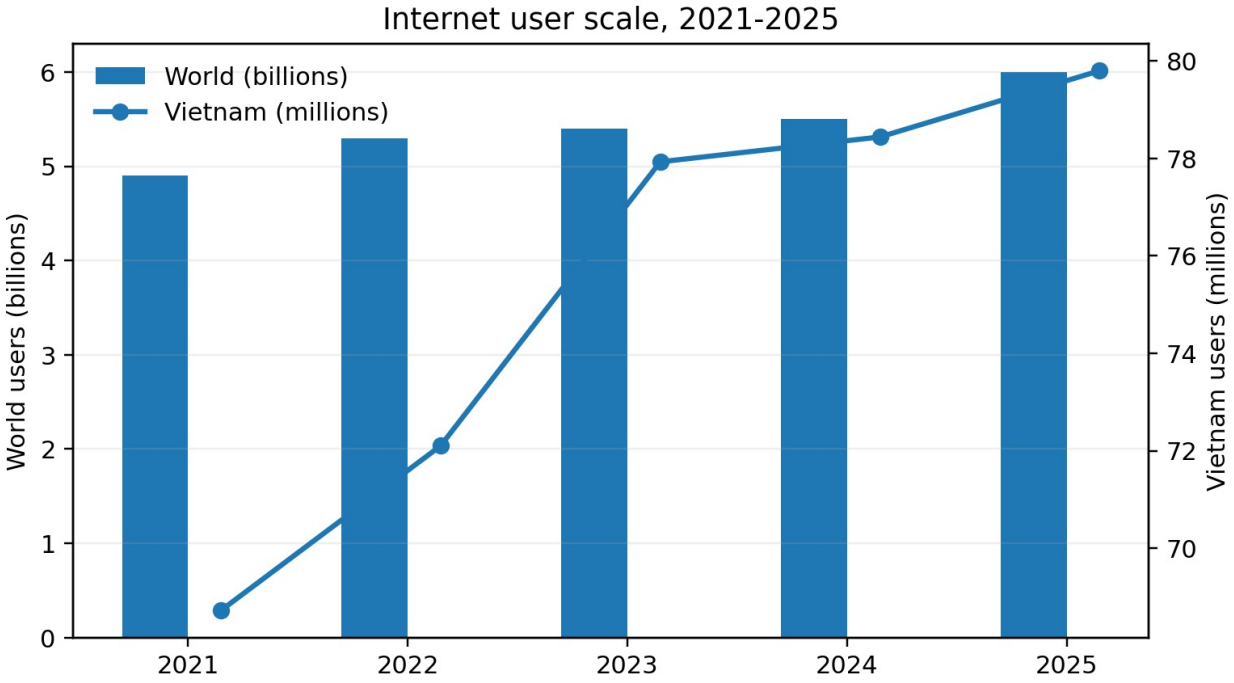


Figure 2. Absolute internet user scale, 2021-2025.

Source: ITU (2021-2025) and DataReportal (2021-2025).

Year	World users (bn)	World penetration	Vietnam users (m)	Vietnam penetration
2021	4.90	63%	68.72	70.3%
2022	5.30	66%	72.10	73.2%

2023	5.40	67%	77.93	79.1%
2024	5.50	68%	78.44	79.1%
2025	6.00	74%	79.80	78.8%

Between 2021 and 2025, Vietnam added 11.08 million internet users (+16.1%), equivalent to a CAGR of 3.8%. The world added about 1.1 billion users (+22.4%), a CAGR of 5.2% using the annual published estimates. The implication for media economics is that audience growth increasingly occurs inside digital environments, raising the value of search, social video, recommendation algorithms and measurable advertising inventory.

Methodological note: ITU revises historical estimates when new information becomes available. For example, its 2025 release revised the 2024 global estimate upward to 5.8 billion users and 71% penetration. Figure 1 uses the figures published in each annual edition to show the information available at each point in time; this is a “publication-vintage” time series rather than a fully revised historical dataset (ITU, 2025).

3. Quantitative Analysis of GAFA Roles in Media and Communication

GAFA is used here as shorthand for Google/Alphabet, Amazon, Facebook/Meta and Apple. Their roles differ: Google dominates search and online video discovery; Meta operates large social and messaging networks; Amazon links advertising to commerce and purchase data; Apple controls a major device, operating-system and app-services ecosystem. Quantitatively, all four expanded monetization between 2021 and 2025.

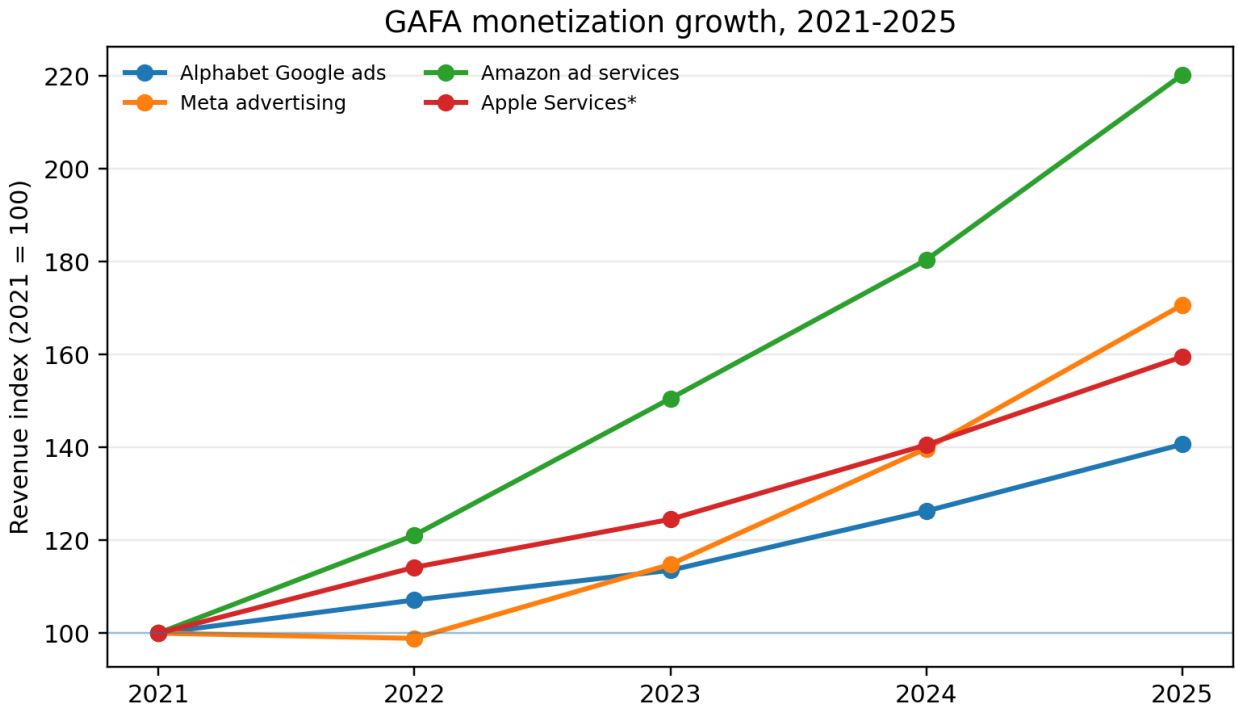


Figure 3. Growth of selected GAFA monetization revenue, indexed to 2021 = 100.

Source: Alphabet, Meta, Amazon and Apple annual reports/financial statements (2022-2026).

Company metric	2021 (US$bn)	2025 (US$bn)	2021-25 change	CAGR
Alphabet Google advertising	209.5	294.7	+40.7%	8.9%
Meta advertising	114.9	196.2	+70.7%	14.3%
Amazon advertising services	31.2	68.6	+120.3%	21.8%
Apple Services*	68.4	109.2	+59.5%	12.4%

Amazon recorded the fastest relative growth in this comparison, more than doubling advertising-services revenue. Meta's advertising revenue rose by about 71%, while Alphabet's much larger advertising base still grew by 41%. Apple Services rose nearly 60%; importantly, this category includes advertising but also AppleCare, cloud services, digital content, payment services and other services, so it should not be interpreted as pure ad revenue (Apple Inc., 2022, 2025).

3.1 Concentration of discovery and audience access

Revenue growth is reinforced by control of audience gateways. StatCounter reported Google at 89.57% of worldwide search-engine usage in May 2025 (StatCounter, 2025). In Vietnam, Google's advertising tools indicated YouTube ad reach of 62.3 million people in early 2025, equal to 78.0% of the country's internet-user base. Meta's tools indicated Facebook ad reach of 76.2 million, equal to 95.4% of the internet-user base (Kemp, 2025). These figures are advertising-reach estimates, not audited active-user counts, but they illustrate the scale available to advertisers through two platform groups.

4. GAFA Market Power: Quantitative Interpretation

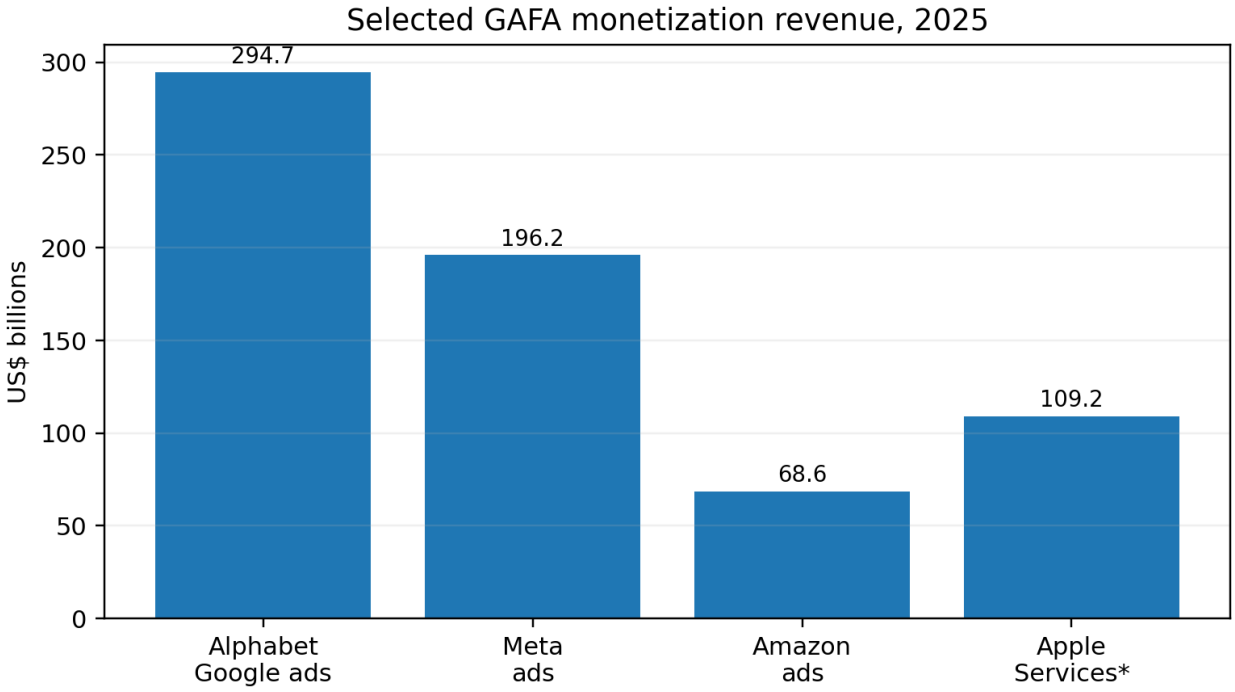


Figure 4. Selected GAFA monetization revenue in 2025.

Source: Company filings. *Apple Services includes non-advertising services; categories are not directly additive.

4.1 Google/Alphabet

Alphabet reported US$294.7 billion in Google advertising revenue in 2025, including search, YouTube ads and Google Network advertising. This scale matters for media because search results and YouTube increasingly function as discovery layers between publishers and audiences. Publishers can gain reach through these platforms but may also become dependent on changes in ranking, recommendation and advertising technology.

4.2 Meta (Facebook, Instagram, Messenger, WhatsApp)

Meta generated US$196.2 billion in advertising revenue in 2025, up 22% year on year. In 2024 its Family daily active people metric averaged 3.35 billion in December, showing the global audience scale that supports ad targeting and content distribution (Meta Platforms, 2025). In Vietnam, Facebook's estimated ad reach was close to the entire measured internet-user base, which gives Meta a major role in news discovery, creator distribution and brand communication.

4.3 Amazon

Amazon advertising services reached US$68.6 billion in 2025. Its distinguishing feature is the connection between media exposure and shopping behavior: sponsored ads, display and video can be targeted within an environment where users search and purchase products. This "retail media" logic expands the definition of media competition because advertising budgets can move from publishers and social platforms toward commerce platforms that offer closed-loop measurement.

4.4 Apple

Apple's 2025 Services revenue reached US$109.2 billion. Apple does not disclose a directly comparable advertising total in the same way as Alphabet, Meta or Amazon. Nevertheless, its devices, operating systems, App Store, privacy rules and service ecosystem influence how content and advertising are distributed and measured. The quantitative importance is the size and growth of the services layer: 59.5% growth from 2021 to 2025.

4.5 Implication for Vietnam

For Vietnamese media organizations, GAFA's role is therefore structural rather than merely competitive. The platforms control large portions of discovery, social distribution, advertising technology, commerce data or app access. This helps explain why the domestic press can experience rising online traffic while still facing weak revenue capture: attention can grow faster than publisher monetization when advertising value is captured upstream by platforms or intermediaries.

5. Quantitative Analysis of Press and Other Media in Vietnam

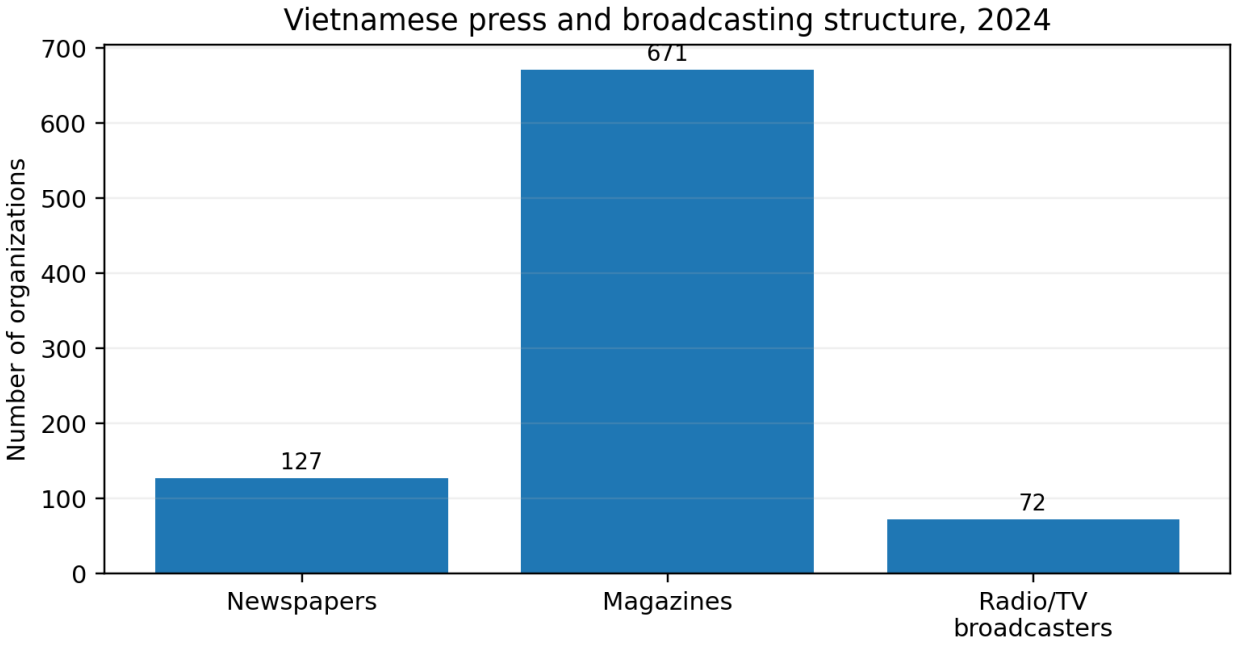


Figure 5. Structure of Vietnamese press and broadcasting organizations, 2024.

Source: Vietnam Ministry of Information and Communications statistics reported by VietNamNet (2024).

By the end of 2024, Vietnam had roughly 884 media/press organizations in the administrative count, including 812 newspapers and magazines and about 72 radio-television broadcasters. A detailed sector summary reported 127 newspapers, 671 magazines and 72 radio/television organizations, with approximately 41,000 people working in journalism and about 21,000 press-card holders (VietNamNet, 2024). Small discrepancies between year-end reports reflect classification and reporting dates rather than a material difference in the market picture.

5.1 Broadcasting and channel supply

The 2024 broadcasting white paper reported that at the end of 2023 Vietnam had 72 radio/television organizations, 78 domestic radio channels, 189 domestic television channels and 45 foreign radio/television channels distributed through pay-TV services. The number of foreign channels fell by 11 from 2022, with regulators noting a shift from linear channels toward on-demand services (VietNamNet, 2024). This is a measurable sign of format substitution: the market is moving from scheduled broadcasting to platform and on-demand viewing.

5.2 Scale of content production

Vietnamese press agencies produce approximately 40 million news items annually, alongside around 50,000 hours of television and 20,000 hours of radio content. Policy-communication content increased from about 11% to 20% of total news output after implementation of the government's policy-communication directive (VietNamNet, 2024). The high output volume means the economic challenge is not scarcity of content; it is how content is discovered, differentiated and monetized in a crowded digital environment.

5.3 "Other media" and digital publishing

Digitalization also appears in publishing. In 2024, Vietnam produced an estimated 4,050 electronic publication titles, up 120.7% year on year, and reported a 200% increase in listening to audiobooks and e-books. Electronic publications accounted for about 8.9% of total publication titles (VietNamNet, 2024). Although publishing is not identical to news media, the direction is consistent with the broader shift toward digital formats and on-demand consumption.

6. Digital Transformation of Vietnamese Press

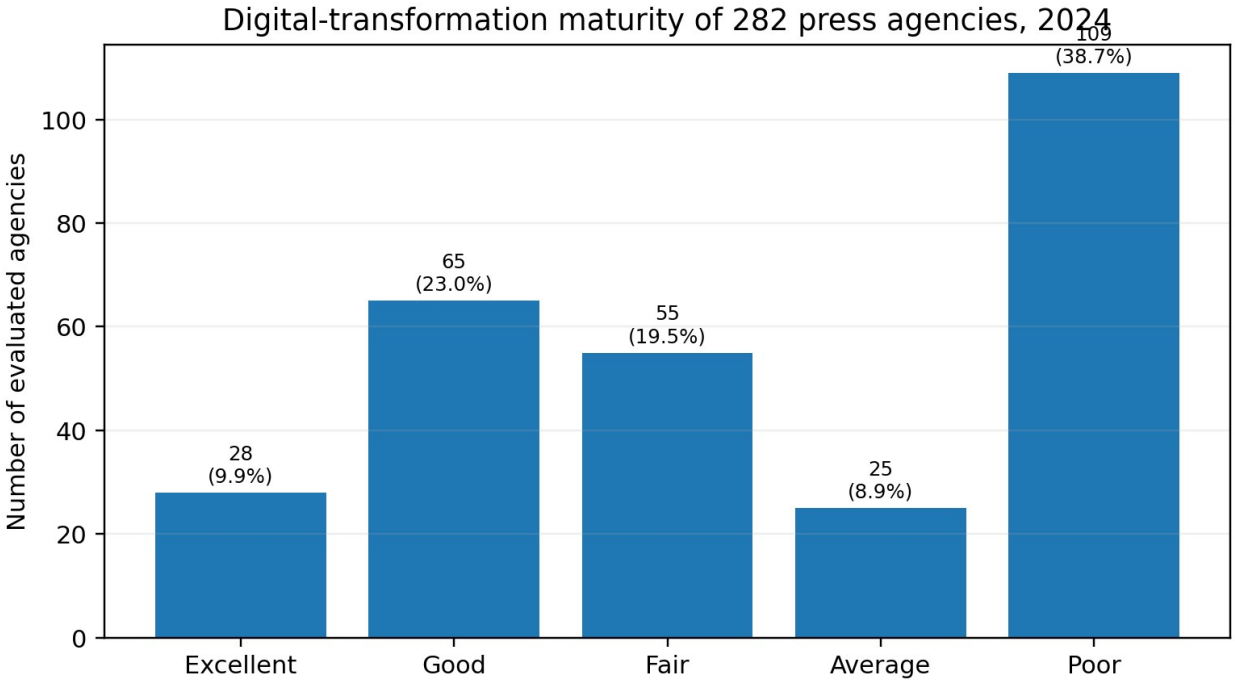


Figure 6. Digital-transformation maturity ratings of Vietnamese press agencies, 2024.

Source: Vietnam Ministry of Information and Communications digital-maturity evaluation, reported by VietNamNet (2024).

In 2024, 282 press agencies participated in the national digital-transformation maturity assessment. Only 28 agencies (9.9%) were rated excellent. A further 65 (23.0%) were good, 55 (19.5%) fair, 25 (8.9%) average, and 109 (38.7%) poor. Thus, 33.0% were good or excellent, while almost two-fifths remained in the lowest category.

6.1 Quantitative meaning of the maturity gap

This distribution shows that digital transformation is real but uneven. The strongest agencies have moved beyond "putting articles online" toward integrated content management, audience analytics, multimedia production, data systems and digital revenue. The Ministry reported that press organizations with successful digital transformation saw traffic growth of roughly 35-40% and around 20% growth in digital-advertising revenue, indicating that capability can affect both audience and monetization outcomes (Ministry of Information and Communications, 2024).

6.2 National targets

Vietnam's press digital-transformation strategy set progressively higher targets for 2023-2025. For example, the share of press agencies operating converged newsrooms and producing content aligned with digital-journalism trends was targeted at 30% in 2023, 50% in 2024 and 80% in 2025. The target for agencies using centralized data analytics and artificial intelligence was 20%, 30% and 50% respectively (Ministry of Information and Communications, 2023). These targets indicate that digital capability has become an explicit performance dimension of the media system.

6.3 Economic interpretation

The maturity data also help reconcile an apparent paradox: online audiences can be large while industry revenues fall. A publisher may gain traffic through search and social media without owning the customer relationship, data or advertising transaction. The commercial advantage therefore shifts toward organizations that can build first-party data, direct subscriptions or memberships, events, branded content, digital services and other revenue streams in addition to conventional display advertising.

7. Quantitative Analysis of the Digital Advertising Market in Vietnam

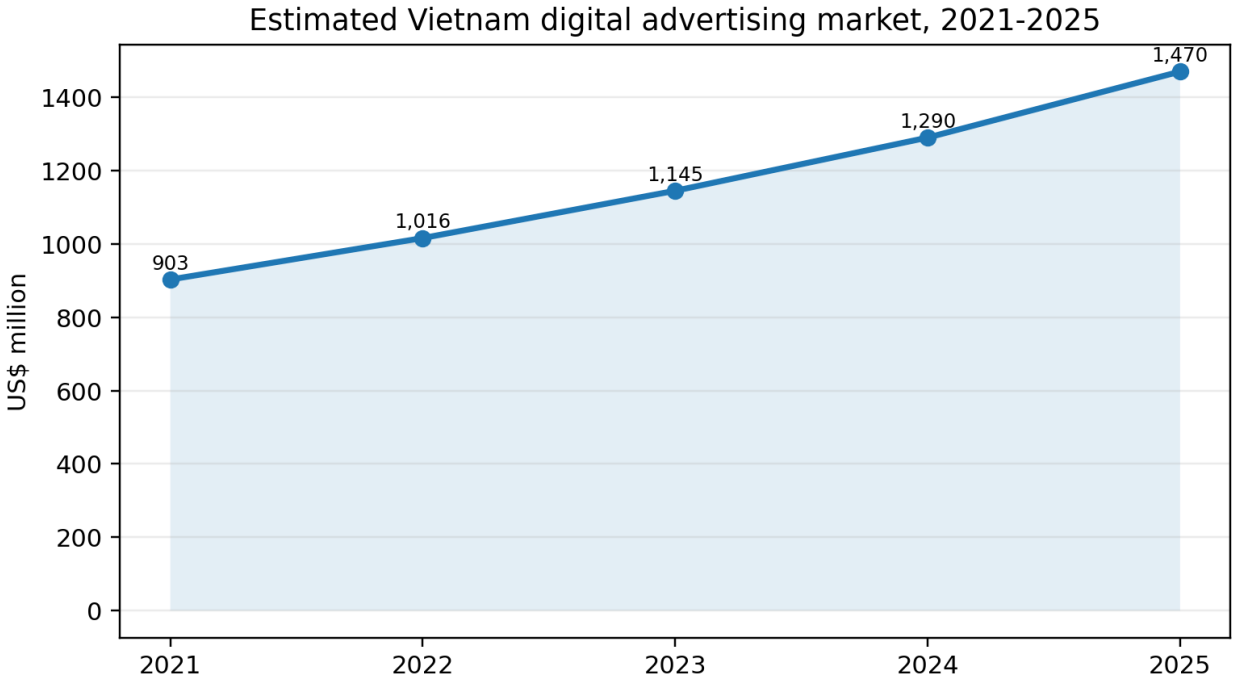


Figure 7. Estimated size of Vietnam's digital advertising market, 2021-2025.

Source: Ken Research (2026) historical/base-year market estimates; 2024 value cross-checked with TechSci Research (2025).

Year	Estimated market size (US$m)	YoY change
2021	903	-
2022	1,016	12.5%
2023	1,145	12.7%
2024	1,290	12.7%
2025	1,470	14.0%

The estimated market expanded from US$903 million in 2021 to US$1.47 billion in 2025, an increase of US$567 million or 62.8%. The implied four-year CAGR is 13.0%. TechSci Research independently estimated the 2024 market at US$1.29 billion and projected a 14.38% CAGR to 2030, which supports the order of magnitude of the series used here (TechSci Research, 2025).

7.1 Drivers visible in the quantitative evidence

- Audience availability: 79.8 million internet users in early 2025 create national-scale addressable reach.
- Social/video reach: YouTube's estimated ad reach was 62.3 million and Facebook's was 76.2 million in early 2025.
- Mobile-first consumption: high smartphone and mobile-broadband adoption make short-form video, social and in-app formats attractive.
- Commerce integration: e-commerce and marketplace advertising allow advertisers to connect impressions with transactions.
- Measurement: digital platforms provide targeting and performance metrics that are harder to reproduce in print, radio or linear television.

7.2 Important caveat

Advertising-market estimates differ across research providers because definitions vary: some include influencer advertising, retail media, connected TV or digital out-of-home, while others do not. The absolute number should therefore be treated as an estimate. The more robust result is the direction and speed of change: multiple sources agree that digital advertising is gaining share and growing faster than the traditional media segments that depend heavily on advertising.

8. Quantitative Analysis of Declining Mass-Media Advertising Income

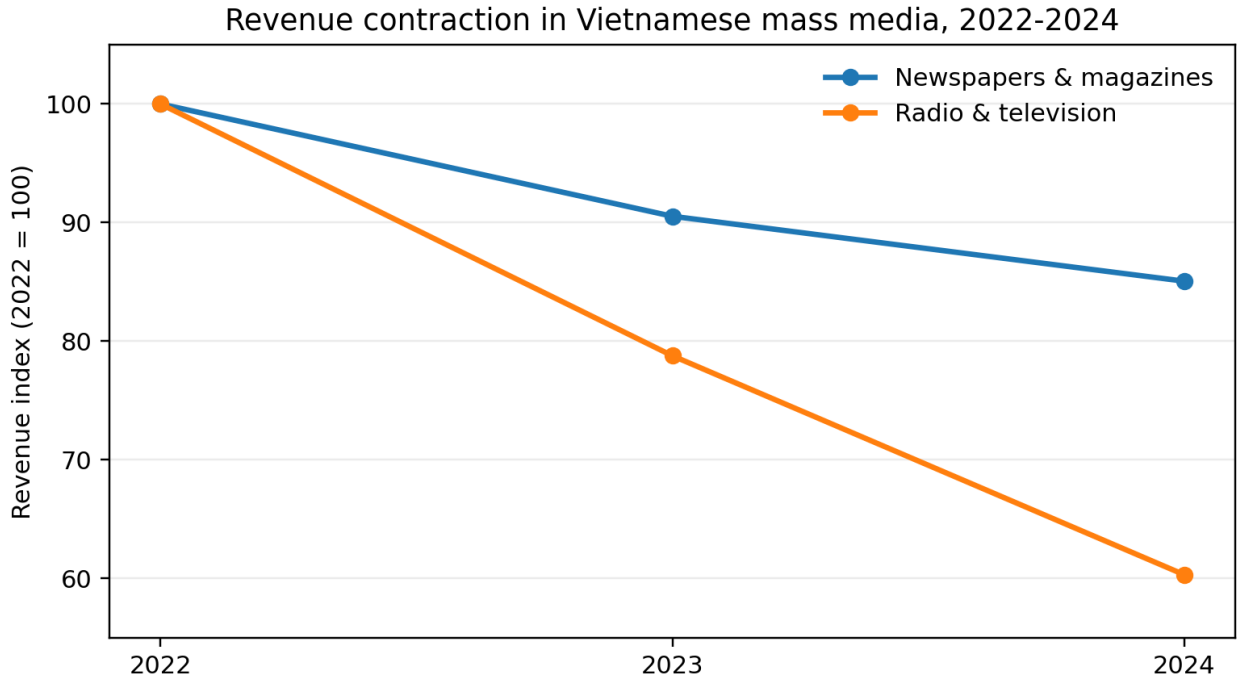


Figure 8. Revenue contraction in newspapers/magazines and radio/television, 2022-2024.

Source: Vietnam Ministry of Information and Communications figures reported by VnEconomy, SGGP and VietNamNet (2024).

Sector	2022 revenue	2023 revenue	2024 revenue	2022-24 change
Newspapers & magazines	VND9,500bn	VND8,600bn	VND8,080bn	-14.9%
Radio & television	VND15,158bn	VND11,939bn	VND9,140bn	-39.7%

The press segment fell from about VND9.5 trillion in 2022 to VND8.6 trillion in 2023 (-9.4%) and VND8.08 trillion in 2024 (-6.1%). The Ministry reported that advertising revenue specifically fell 14.8% in 2023 and another 5.6% in 2024. Because the reported 2022 revenue component combines advertising and circulation in some summaries, this report uses the official percentage changes rather than constructing a potentially misleading nominal advertising series (VnEconomy, 2024; VietNamNet, 2024).

Broadcasting experienced a larger contraction: total radio and television revenue fell from about VND15.16 trillion in 2022 to VND11.94 trillion in 2023 and VND9.14 trillion in 2024. That is a cumulative decline of approximately 39.7% in two years. Regulatory and industry reports attribute the pressure partly to competition from social networks, video platforms and other digital media for commercial advertising (VietNamNet, 2024).

8.1 Why advertising income is vulnerable

Traditional media historically relied heavily on advertising; industry discussions report advertising shares of 60-90% of revenue for some publishers. When advertisers migrate to platforms offering granular targeting, rapid optimization and conversion measurement, publishers can lose revenue even if journalism consumption remains high. This produces the central economic challenge of digital journalism: audience attention can expand while the unit value captured by the publisher declines (Vietnam News, 2024).

8.2 Not all decline should be attributed to platforms

Causality should be stated carefully. Revenue changes also reflect macroeconomic conditions, agency restructuring, post-COVID normalization, changes in state funding, production costs and audience behavior. The quantitative evidence establishes simultaneous digital-ad growth and legacy-media revenue decline; it supports a structural-shift interpretation but does not prove that every lost dong of traditional-media revenue moved directly to a foreign platform.

9. Five-Year Synthesis: How Rapidly Is Vietnam Changing?

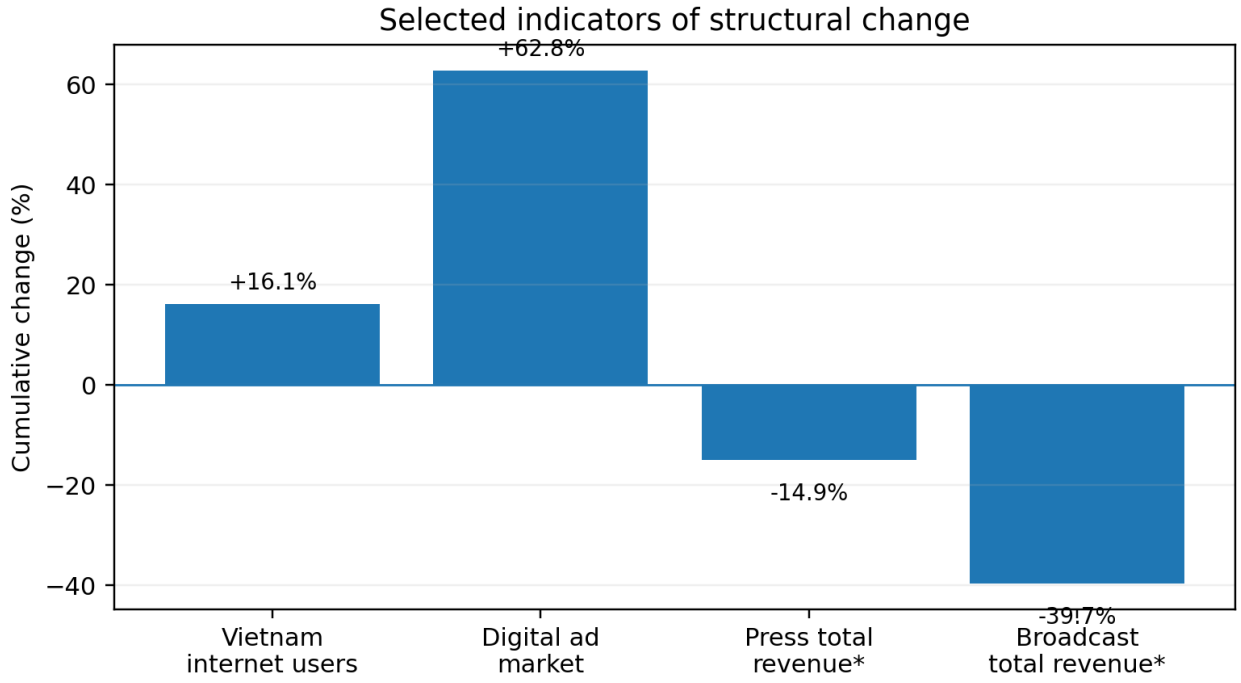


Figure 9. Selected indicators of structural change in Vietnam's media market.

Source: Author calculations from sources cited in Sections 2, 7 and 8. *Traditional-media revenue comparison uses 2022-2024 because comparable official figures are available for those years.

9.1 A quantitative "direction-of-change" test

The strongest way to test the statement "media and communication is changing rapidly in Vietnam" is to examine whether independent indicators move in a consistent structural direction. They do. Internet users rose by 16.1% between 2021 and 2025, while the digital advertising market estimate rose 62.8%. In contrast, newspapers/magazines lost 14.9% of total revenue from 2022 to 2024 and broadcasters lost 39.7%. This divergence is too large to describe as a small channel adjustment.

Dimension	Quantitative evidence	Interpretation
Access	79.8m internet users in 2025	Digital is a mass medium, not a niche
Platform reach	Facebook 76.2m; YouTube 62.3m ad reach	Foreign platforms can access national-scale audiences
Advertising	Digital ad market +62.8% (2021-25 est.)	Budgets are shifting toward measurable digital channels
Press economics	Press revenue -14.9% (2022-24)	Legacy monetization is under pressure
Broadcast economics	Radio/TV revenue -39.7% (2022-24)	Linear media faces stronger revenue disruption
Capability	38.7% of assessed agencies rated poor in digital maturity	Transformation is rapid but uneven

9.2 Pace versus maturity

The market is changing faster than many organizations can transform. The digital-advertising CAGR of about 13% is much higher than Vietnam's internet-user CAGR of about 3.8%, implying that monetization is deepening faster than connectivity itself. In other words, the next phase is not mainly about adding first-time internet users; it is about increasing time spent, video consumption, commerce integration, targeting sophistication and advertiser spending per connected user.

9.3 Structural conclusion

The five-year evidence supports a structural-change interpretation: digital platforms have become primary gateways for attention and advertising; traditional media remains important for public information and journalism but must operate across digital distribution systems; and revenue models are being diversified because traditional advertising alone is increasingly insufficient.

10. Discussion: Implications for Media and Communication Markets

10.1 From channel competition to ecosystem competition

The relevant competitive unit is no longer simply newspaper versus television versus radio. Digital ecosystems bundle discovery, identity, messaging, video, cloud services, app distribution, commerce and advertising. GAFA's revenue growth quantifies the scale of this ecosystem logic. A local publisher competes not only for "readers" but for time, data, advertiser budgets and direct customer relationships.

10.2 Platform dependence and bargaining power

When Facebook or YouTube can offer reach equivalent to a very large share of Vietnam's internet population, advertisers may buy reach directly from platforms. Publishers still supply socially valuable content, but platforms can intermediate discovery and advertising. This creates a bargaining-power asymmetry because changes in algorithms, privacy policies or advertising technology can change publisher traffic or yield without a corresponding decision by the publisher.

10.3 The likely direction of sustainable media economics

The quantitative evidence suggests that Vietnamese media organizations will increasingly combine multiple revenue streams: digital advertising, subscriptions or memberships, events, branded content, licensing, data services, e-commerce/affiliate activity, technology services and public-service funding where applicable. The 2024 revenue contraction strengthens the case for diversification, while the traffic and digital-ad gains reported among digitally mature organizations show that transformation can improve economic outcomes.

10.4 Measurement priorities for future research

A stronger final research project could collect outlet-level data on website traffic, social referrals, direct traffic, video views, digital subscriptions, advertising revenue, newsroom size and digital-maturity scores. Regression analysis could then test whether digital maturity predicts revenue growth after controlling for outlet size and type. A second approach would use interrupted time series to compare pre- and post-transformation performance at a specific media organization.

10.5 Suggested hypotheses

- H1: Higher digital-maturity scores are positively associated with digital advertising revenue growth among Vietnamese press agencies.
- H2: The share of audience traffic arriving through social/search platforms is negatively associated with direct audience ownership, measured by logged-in users or subscribers.
- H3: Revenue diversification is associated with lower volatility in total newsroom revenue.
- H4: Video-first and mobile-first publishing is associated with higher reach among younger audiences.

11. Conclusion and Recommendations

This report finds strong quantitative evidence that the media and communication market has changed rapidly in Vietnam over the last five years. The most important change is a divergence between digital scale and legacy-media economics. Vietnam's internet audience reached nearly 80 million by early 2025; estimated digital advertising grew by almost 63% from 2021 to 2025; Facebook and YouTube offered national-scale advertising reach; and the world's largest digital platforms increased monetization substantially. Meanwhile, Vietnamese press and broadcasting revenues declined sharply in 2022-2024.

The evidence does not imply that journalism or traditional media has become irrelevant. Vietnam still has a large institutional press system, high content output and a substantial broadcasting sector. Rather, the market logic has changed: distribution is increasingly platform-mediated, advertising is increasingly data-driven and measurable, and sustainable media economics requires stronger digital capability and more diversified revenue.

Recommendations for the final graded assessment

Recommendation	How to apply it
Use a clear dependent variable	Choose revenue, ad revenue, audience reach, or digital-maturity score rather than "change" in general.
Keep time periods comparable	Use the same years and definitions when calculating growth rates; flag breaks in methodology.
Separate official data from estimates	Label commercial market sizes as estimates and avoid presenting them as government statistics.
Distinguish correlation from causation	Digital ad growth and press decline are simultaneous, but not every decline is caused only by platforms.
Use multiple indicators	Combine access, platform reach, market revenue and organizational capability for a stronger argument.
Add an original dataset if possible	Outlet-level observations can support correlation or regression and improve the academic contribution.

Final statement

Quantitatively, Vietnam's media transformation is characterized by high and still-growing connectivity, rapidly expanding digital advertising, very large platform audiences, declining traditional-media revenue, and uneven but accelerating newsroom digital transformation. These patterns are mutually reinforcing and provide a strong empirical background for further research on platformization, advertising economics and the future of journalism in Vietnam.

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Appendix A. Core Data Used in the Calculations

A1. Internet users

Year	World users (bn)	World penetration	Vietnam users (m)	Vietnam penetration
2021	4.90	63%	68.72	70.3%
2022	5.30	66%	72.10	73.2%
2023	5.40	67%	77.93	79.1%
2024	5.50	68%	78.44	79.1%
2025	6.00	74%	79.80	78.8%

A2. GAFA selected monetization metrics (US$ billions)

Year	Alphabet ads	Meta ads	Amazon ads	Apple Services*
2021	209.497	114.934	31.160	68.425
2022	224.473	113.642	37.739	78.129
2023	237.855	131.948	46.906	85.200
2024	264.590	160.633	56.214	96.169
2025	294.691	196.175	68.635	109.158

*Apple Services is not directly comparable with the other three advertising categories because it includes advertising, AppleCare, cloud services, digital content, payment services and other services.

A3. Example calculations

Vietnam internet-user growth = (79.80 / 68.72 - 1) x 100 = 16.1%.

Vietnam internet-user CAGR = (79.80 / 68.72)^(1/4) - 1 = 3.8% per year.

Vietnam digital-advertising CAGR = (1,470 / 903)^(1/4) - 1 = 13.0% per year.

Broadcast revenue change, 2022-2024 = (9,140 / 15,158 - 1) x 100 = -39.7%.

A4. Interpretation rules

- Do not treat advertising reach as identical to monthly active users.
- Do not add Apple Services to advertising revenues as if all components were advertising.
- Do not compare commercial market estimates with official administrative revenue as if they used the same accounting definition.
- Use rates and indexes when definitions are stable but nominal baselines are not fully disclosed.